REAL-TIME STREAMING ANALYTICS PIPELINE IN AWS FOR HEALTH MONITORING

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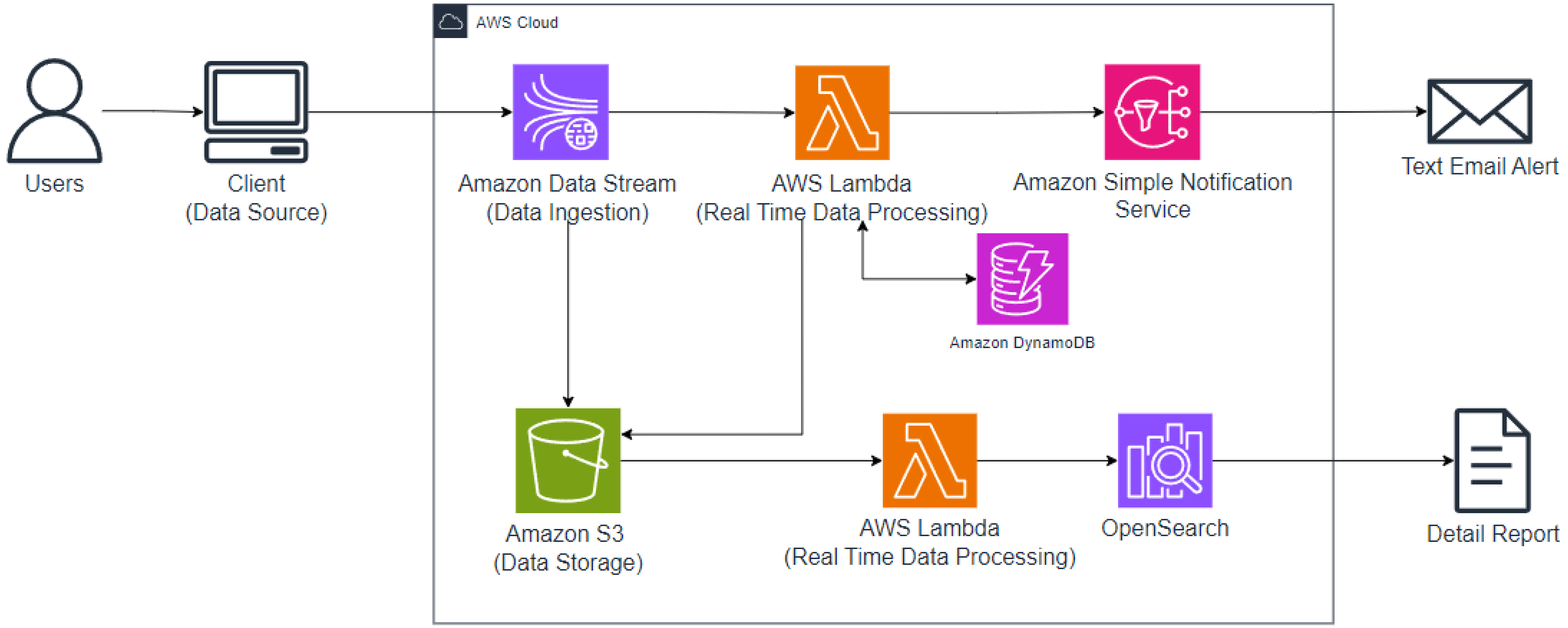
Introduction

- Real-time streaming analytics allows for continuous processing of data streams, enabling timely decision-making and insights.
- Leverage AWS services to build a scalable, secure, and efficient analytics pipeline for monitoring real-time health data.

Introduction

- **Why It Matters?**:
  - Immediate detection of health anomalies.
  - Enhance healthcare outcomes.
- **Key Focus Area:**
  - AWS Services (Amazon Kinesis, Amazon DynamoDB, SNS, Lamda, S3)
  - Data ingestion, processing, and analysis.
  - Real time alert and dashboards(Amazon Open Search)


AWS Cloud
Users
Client
(Data Source)
Amazon Data Stream
(Data Ingestion)
AWS Lambda
(Real Time Data Processing)
Amazon Simple Notification
Service
Text Email Alert
Amazon DynamoDB
Amazon S3
(Data Storage)
AWS Lambda
(Real Time Data Processing)
OpenSearch
Detail Report

Methodology

- **Data Ingestion**
- **Amazon Kinesis** is used to **capture real-time health data** streams. It enables continuous data ingestion from sources like IoT devices (e.g., medical sensors).
- **How it works**:
- Kinesis ingests a high volume of data in real time, splitting the data into shards for efficient processing.
- **Provisioned mode** can be configured with **1 or more shards**, allowing scalable data capture and processing.

Methodology

- **Cold Storage**
- **Amazon Kinesis Firehose** delivers raw data from **Kinesis Stream** to **Amazon S3** for long-term storage.
- Data from Kinesis Stream is compressed in **GZIP** format using Firehose and stored in S3 for later analysis.
- **Amazon S3** stores large amounts of data with **high durability and availability**, making it ideal for archiving raw health data.
- **Scalable** storage with integrated **lifecycle policies** to manage data cost-effectively.

Methodology

- **Hot Data**
- **Kinesis Data Analytics** processes real-time data for anomalies (e.g., seizure detection) using SQL-based stream analytics.
- **AWS Lambda** is triggered when a threshold is breached. The Lambda function:
- Sends real-time alerts to stakeholders via **Amazon SNS**.
- Writes anomaly details to **Amazon DynamoDB** for further processing and analysis.
- **DynamoDB Table**: Stores patient information, including event timestamps and details of seizures.

Results And Outcomes

- **Data Storage and Management:**

REAL TIME SEIZURES TABLE

age	currentLocation	eventTimestamp	heartRate	hospitalId	inspectedAt	name
82	null	2024-08-19 12:50:43.178734	161	null	2024-08-19 07:05:44.649295	James Thompson
61	null	2024-08-19 12:50:27.975594	80	null	2024-08-19 07:05:29.679300	Tina Johnson
67	null	2024-08-19 12:50:26.383464	76	null	2024-08-19 07:05:28.668891	Stephen Fuller
74	null	2024-08-19 12:49:58.911034	161	null	2024-08-19 07:05:00.665144	Christian Foster
85	null	2024-08-19 12:49:51.711964	150	null	2024-08-19 07:04:53.632263	Bryan Frank

Methodology

- **Amazon Simple Notification Service (SNS)** sends real-time alerts (e.g., email or SMS) to healthcare providers when critical health events like seizures are detected.
- **How it works**:
- SNS subscribes healthcare personnel to specific alerts.
- When a **Lambda function** triggers SNS, it broadcasts notifications to all subscribers.
- It ensures that **real-time notifications** are delivered instantly, improving response times in emergencies.

Results And Outcomes

- **Real-Time Notification:**

AWS Notification Message Inbox ×

Seizure Notification <no-reply@sns.amazonaws.com>
to me

12:44 PM (5 hours ago)

🚨 Seizure Alert 🚨

Patient with seizure detected!

Patient Name: Samantha Allen
Patient ID: eb6d1bf2-ef4e-4a06-82d9-1fe1da9e2305
Age: 55
Heart Rate: 107 BPM
Respiratory Rate: 18 breaths/min
Oxygen Saturation: 92%
Seizure Severity: Severe
Seizure Duration: 178 seconds
Location: null
Event Timestamp: 2024-08-19 12:44:35.320600

Please take immediate action.

Methodology

- **Amazon OpenSearch Service** is used to **visualize real-time health data** on customizable dashboards.
- **How it works**:
- Data is indexed in **OpenSearch** via **Lambda functions**, enabling dynamic querying and visualization.
- Dashboards are created for monitoring **patient vitals**, **anomalies**, and trends like seizure occurrence.
- It allows powerful **visualization tools** (e.g., bar charts, pie charts) to display complex datasets, making it easy for healthcare teams to monitor patient health.

Results And Outcomes

- **Data Visualization Using OpenSearch:**

Methodology

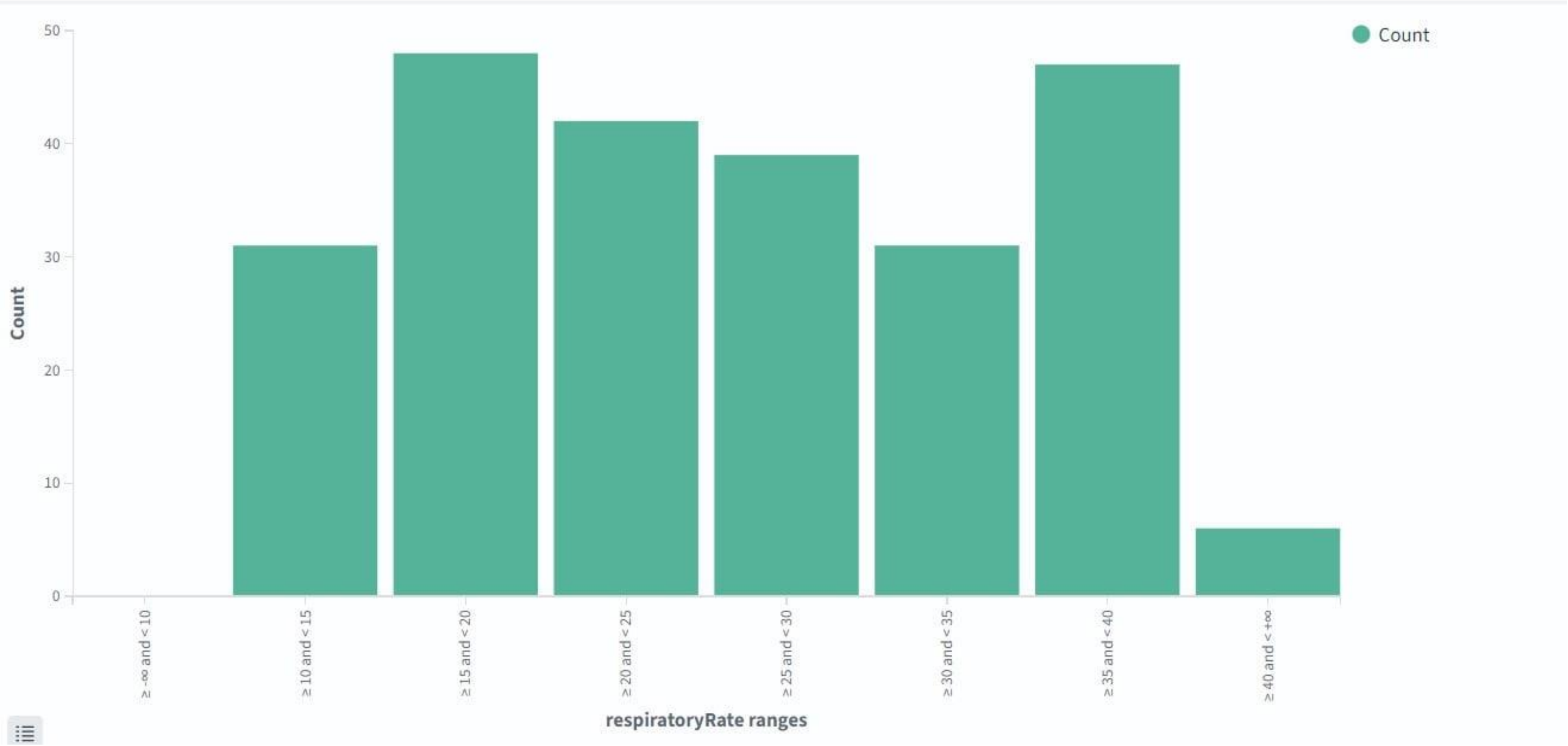
- **AWS Cloud Development Kit (CDK)** allows defining cloud infrastructure using code.
- It simplifies **cloud deployment** and offers a **high-level abstraction** using familiar programming languages.
- **Benefits:**
  - **Infrastructure as Code (IaC)** for easier **replication** and scalability.
  - Integration with multiple AWS services.
  - **Automated** health monitoring system setup.
  - **Customizable pipelines** for different health monitoring use cases.

Methodology

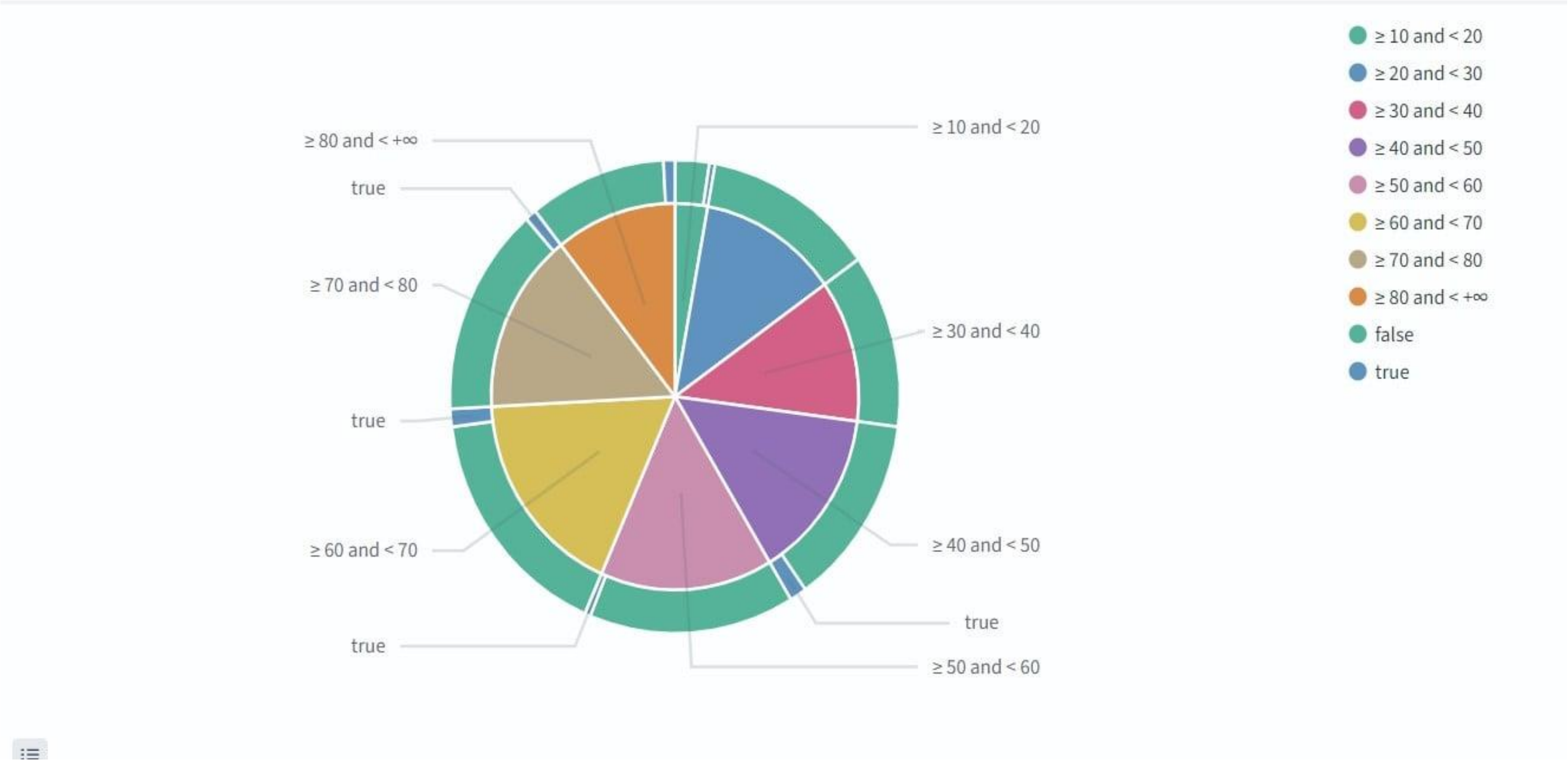
```
// ==============================================================================
// OpenSearch Module:

const opensearchDomain = new Domain(this, 'Domain', {
  version: EngineVersion.OPENSEARCH_2_13,
  domainName: 'opensearch-domain',
  capacity: {
    dataNodes: 1,
    dataNodeInstanceType: 't3.medium.search',
    multiAzWithStandbyEnabled: false,
  },
  ebs: {
    volumeSize: 20,
  },
  enforceHttps: true,
    nodeToNodeEncryption: true,
    encryptionAtRest: {
      enabled: true,
    },
  fineGrainedAccessControl: {
    masterUserName: process.env.OPENSEARCH_USER,
    masterUserPassword: cdk.SecretValue.unsafePlainText(process.env.OPENSEARCH_PASSWORD as string),
  },
  accessPolicies: [
    new iam.PolicyStatement({
      actions: ['es:*'],
      resources: ['*'],
      effect: iam.Effect.ALLOW,
      principals: [new iam.AnyPrincipal()],
    })],
    removalPolicy: cdk.RemovalPolicy.DESTROY,
});
```

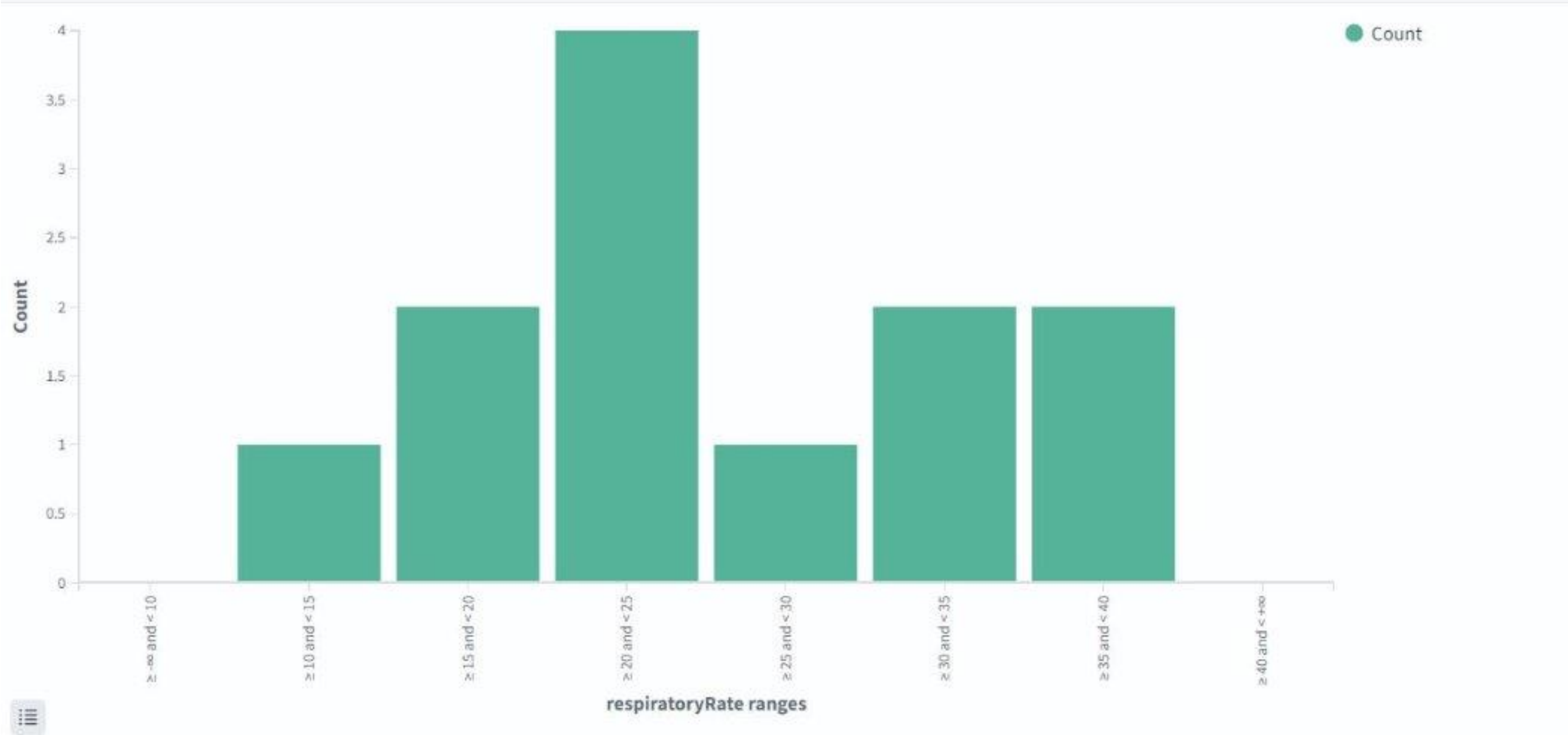
Results And Outcomes



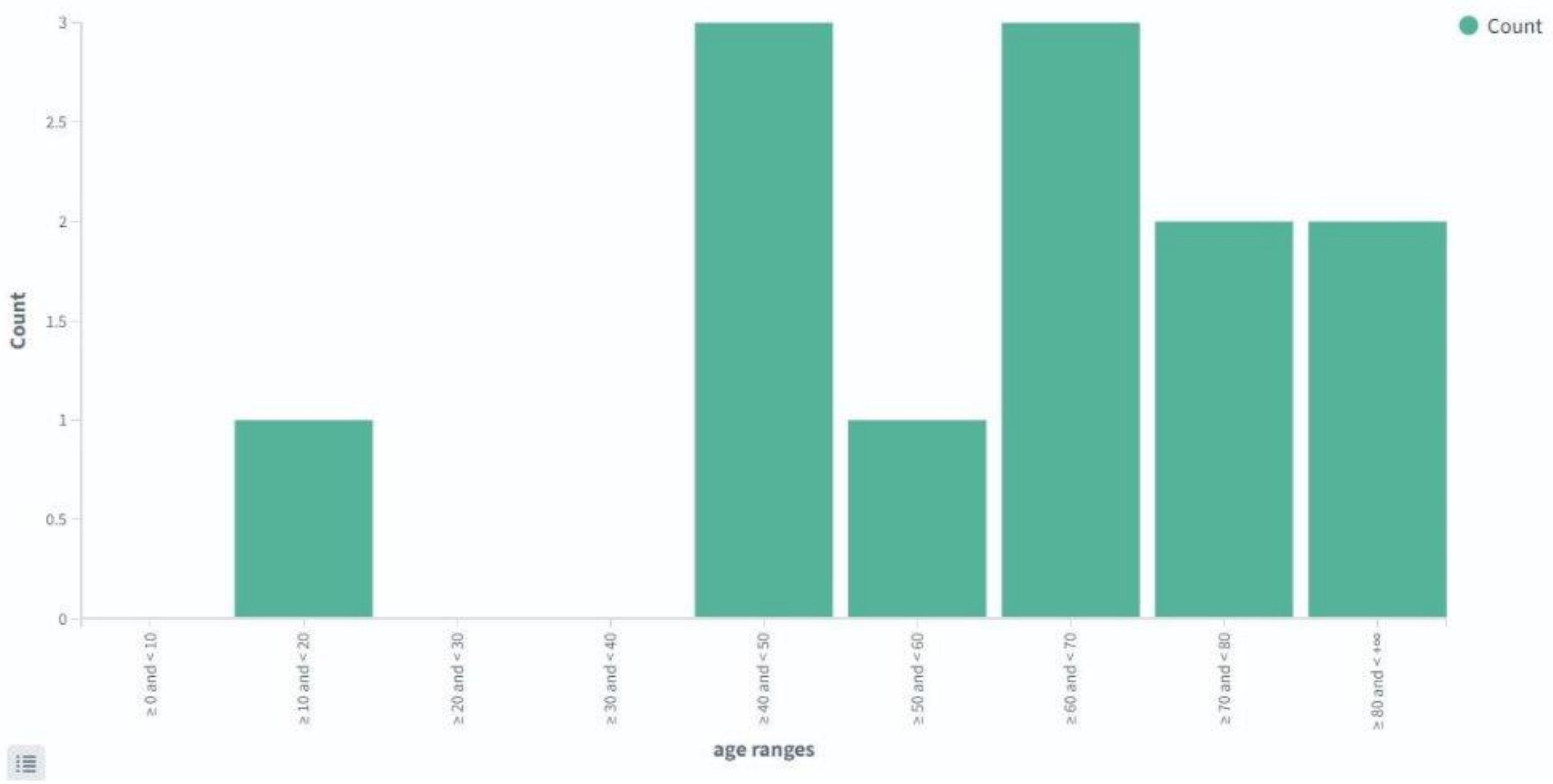
Results And Outcomes



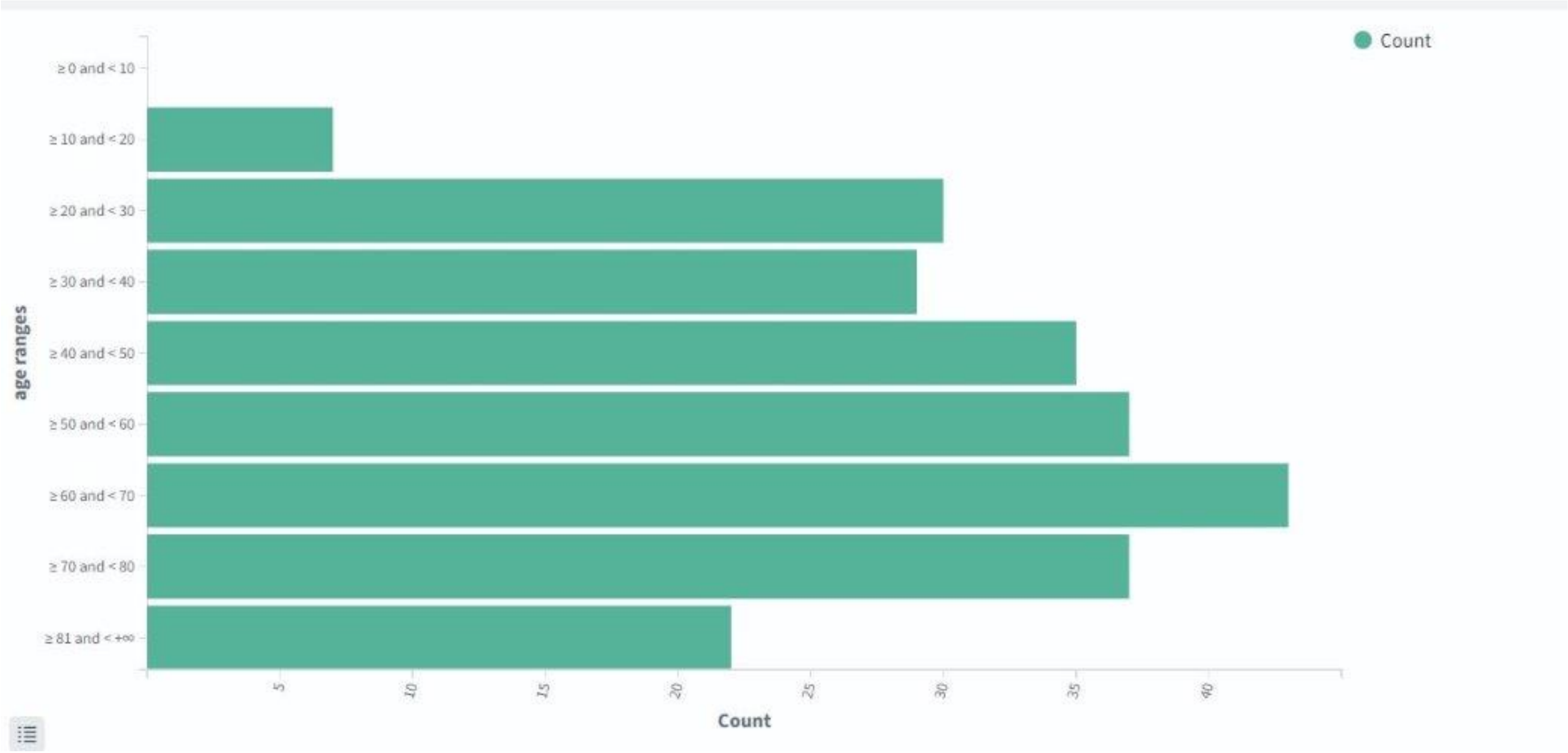
Results And Outcomes



Results And Outcomes

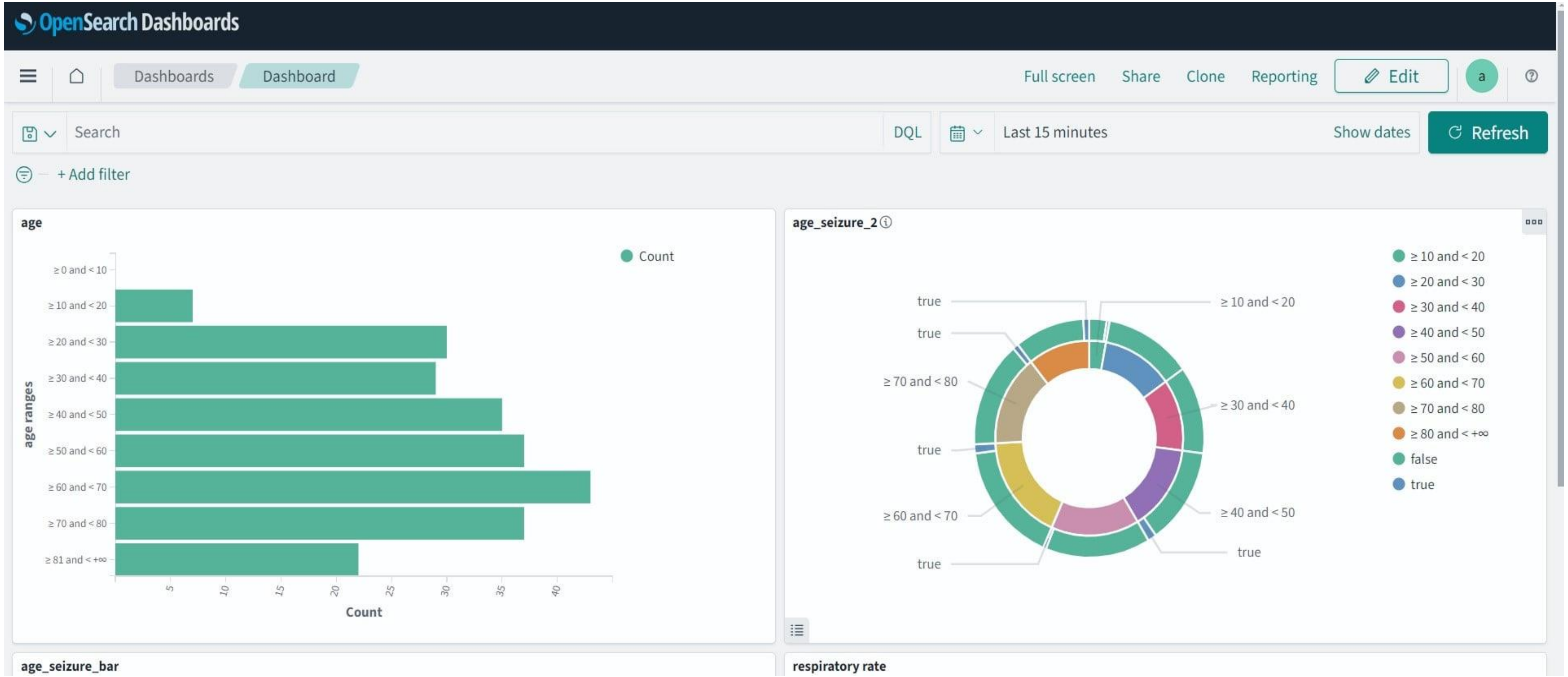


Results And Outcomes



Results And Outcomes

- **Dashboard Creation:**



Results And Outcomes

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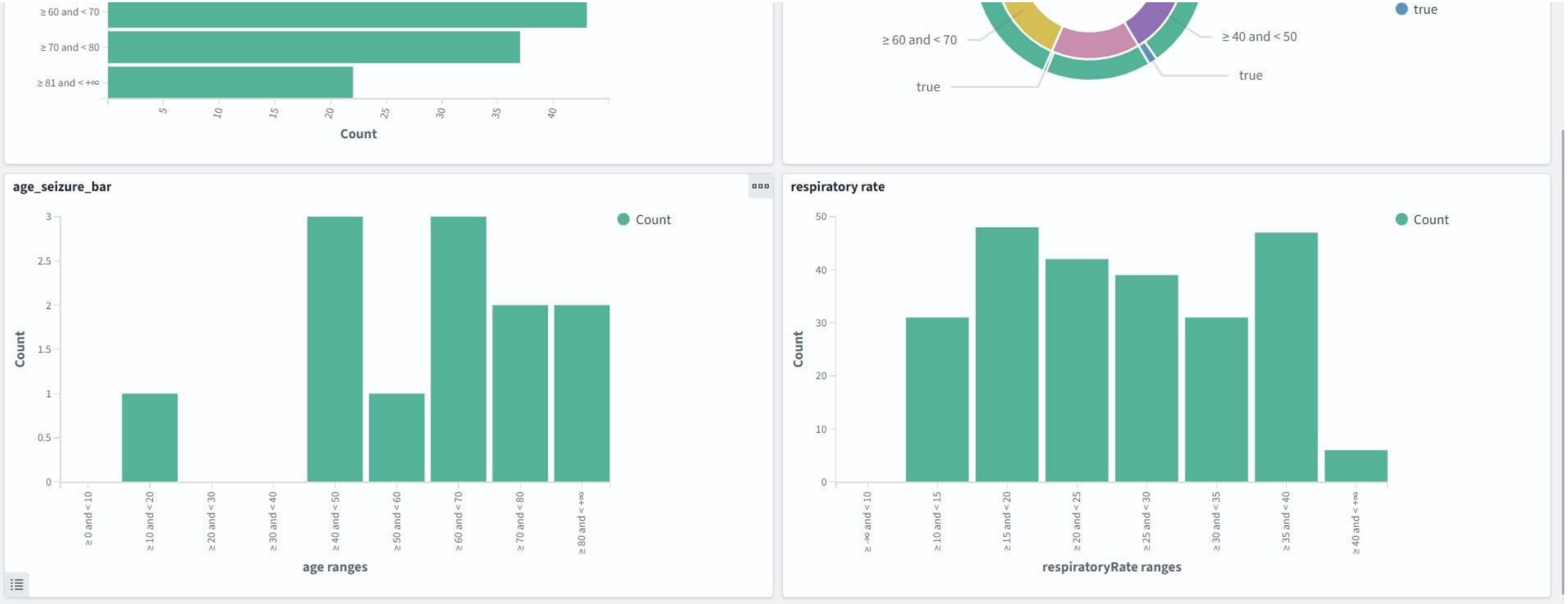

DEMO

Conclusion

- The project provides **Continuous real-time data ingestion** from wearable devices and medical sensors along with **Timely alerts** and notifications via SNS for rapid response to health changes. Also provides **Secure and scalable storage** in S3 for both real-time and historical data and OpenSearch for **Visualization**.
- Enhances **patient outcomes** and supports **personalized treatment plans**. **Optimizes healthcare resources** and improves overall healthcare efficiency.

Thank You